

MINI PROJECT SYNOPSIS

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**Project Title: Dual-Stream Spatial and Frequency-Domain Deepfake Video
Detection using EfficientNet-B4 and Discrete Cosine Transform (DCT)**

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1. Introduction

Deepfake technology refers to the use of deep learning — particularly Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) — to synthesize hyper-realistic video content in which a person's face is digitally replaced or manipulated. The term "deepfake" combines "deep learning" and "fake," and these AI-generated videos have become increasingly indistinguishable from authentic recordings, posing severe threats to digital identity verification, political integrity, and financial security.

The proposed project, DeepGuard AI, addresses this problem by developing a Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection System. Unlike conventional detectors that rely solely on pixel-level (spatial) analysis, our system simultaneously analyzes facial images in two complementary domains: (1) the spatial domain using an EfficientNet-B4 convolutional backbone, and (2) the frequency domain using a 2D Discrete Cosine Transform (DCT) spectral pipeline. The outputs of both streams are fused using an attention-gated mechanism to produce a final authenticity verdict, accompanied by Grad-CAM visual explanations and an interactive Streamlit web dashboard.

Technology Domain

This project falls under the domains of Computer Vision, Deep Learning, and Digital Forensics. It leverages state-of-the-art neural network architectures, signal processing mathematics, and explainable AI to address a real-world cybersecurity problem.

Key Technical Terms

- **Deepfake:** An AI-generated video in which a person's facial appearance is manipulated or replaced using neural network models.
- **GAN (Generative Adversarial Network):** A neural network framework consisting of a generator (that creates fake images) and a discriminator (that identifies fakes), trained adversarially.
- **2D-DCT (Discrete Cosine Transform):** A mathematical transformation that converts an image from the spatial pixel domain into frequency components, revealing hidden periodic artifacts left by GAN upsampling.
- **EfficientNet-B4:** A compact, high-accuracy convolutional neural network by Google that uses compound scaling of depth, width, and resolution for optimal performance-to-size ratio.
- **Grad-CAM (Gradient-weighted Class Activation Map):** An explainability technique that generates a visual heatmap showing which facial regions the neural network focused on when making its classification decision.
- **Attention-Gated Fusion:** A learnable mechanism that dynamically assigns weights to spatial and frequency features based on their discriminative relevance for a given input.

The system is trained and evaluated on standard benchmark datasets — FaceForensics++ (FF++) and Celeb-DF v2 — and deployed via a Streamlit web interface, enabling forensic examiners, security researchers, and verification platforms to upload videos and receive real-time detection results with visual evidence.

2. Statement of Problem and Objectives

2.1 Statement of Problem

The exponential proliferation of AI-generated deepfake videos poses critical threats across multiple sectors: digital banking KYC bypass, political disinformation, and judicial evidence manipulation. Existing deepfake detection systems exhibit the following major limitations:

- **Spatial-Only Vulnerability:** Conventional RGB-based CNN detectors (e.g., XceptionNet, MesoNet) achieve high accuracy on uncompressed videos but lose 20–35% accuracy when videos are compressed by platforms like WhatsApp or YouTube (H.264 codec), because compression destroys subtle pixel-level manipulation artifacts.
- **Lack of Frequency-Domain Analysis:** No mainstream detection system exploits the systematic spectral fingerprints that GAN upsampling operations leave in the frequency domain — fingerprints that remain detectable even after heavy compression.
- **Forensic Opacity:** Most deployed systems output only a probability score without any explanation of which facial regions triggered the detection, making them inadmissible in legal and regulatory contexts.
- **Deployment Gap:** Research-grade detection systems exist only as command-line scripts with no user-friendly interface for non-expert forensic examiners or banking officers.

2.2 Objectives

1. To implement automated facial region extraction from continuous video streams using MTCNN face detection with 15% contextual boundary margins.
2. To design and implement a dual-stream deep learning architecture that processes facial images concurrently in the spatial domain (EfficientNet-B4) and the frequency domain (2D-DCT + Spectral CNN).
3. To formulate and implement the Radial Azimuthal DCT Spectrum $A(r) = (1/|S_r|) \sum |F(u,v)|$ to extract rotationally invariant GAN upsampling fingerprints from frequency maps.
4. To train and rigorously evaluate the dual-stream model on FaceForensics++ (c23 and c40 compression tiers) and Celeb-DF v2 datasets, targeting $AUC-ROC \geq 0.968$ and detection accuracy $\geq 94\%$ under compression.
5. To implement Gradient-weighted Class Activation Mapping (Grad-CAM) for forensic explainability, generating spatial heatmaps that visually indicate which facial regions triggered the detection verdict.
6. To develop and deploy an interactive forensic web dashboard using Streamlit with real-time video upload, frame-by-frame analysis, confidence gauge visualization, and DCT spectrum display.

3. Literature Survey

A systematic review of prior work identifies both the progress and critical gaps in deepfake detection research:

- Rossler et al. (2019) — FaceForensics++: Introduced the FaceForensics++ benchmark with 5,000 videos across four manipulation methods. Evaluated XceptionNet as baseline, achieving 98.5% on raw but only 89.3% on c23-compressed videos, demonstrating the compression vulnerability of spatial-only detectors.
- Frank et al. (2020) — Frequency Analysis: Demonstrated that GAN-generated images exhibit systematic spectral discrepancies in the Fourier domain. This foundational work established that frequency-domain fingerprints of GAN upsampling persist even after compression — the core theoretical basis of our frequency stream.
- Qian et al. (2020) — F3-Net: Proposed the Frequency in Face Forgery Network using DCT-based frequency-aware clues. Achieved 90.4% on c23-compressed FF++, outperforming spatial-only methods, but lacks spatial semantic features and explainability.
- Afchar et al. (2018) — MesoNet: Designed a compact mesoscopic CNN (MesoNet-4) for real-time detection. Lightweight but limited representational capacity, failing on modern high-resolution deepfakes (70.5% on c23).
- Li et al. (2020) — Celeb-DF: Introduced the Celeb-DF v2 dataset with 5,639 high-quality celebrity deepfakes exhibiting minimal visual artifacts. Revealed that models trained on FF++ generalize poorly to cross-dataset evaluation.
- Tan & Le (2019) — EfficientNet: Proposed compound scaling of CNN depth, width, and resolution. EfficientNet-B4 achieves ImageNet accuracy competitive with larger models at $3\times$ fewer parameters, making it ideal as our spatial backbone.
- Selvaraju et al. (2020) — Grad-CAM: Introduced gradient-weighted class activation mapping for visual explanations from CNNs. Applied in our system to identify facial regions that trigger deepfake classification, enabling forensic transparency.
- Durall et al. (2020) — Spectral Analysis of GANs: Showed that CNN-based generative models fail to reproduce natural image spectral distributions, with upsampling operations introducing periodic checkerboard artifacts in DCT spectra — directly exploited in our frequency stream.

4. Expected Outcome / Scope of the Project

4.1 Expected Outcomes

1. A trained PyTorch model (DualStreamNet) achieving $\geq 98.9\%$ detection accuracy on uncompressed FaceForensics++ and $\geq 94.2\%$ on c23-compressed videos, significantly outperforming spatial-only baselines.
2. A functional Streamlit web dashboard allowing upload of video files, automatic face extraction, real-time per-frame classification with confidence scores, DCT spectrum visualization, and Grad-CAM heatmap generation.
3. Grad-CAM explainability heatmaps that visually identify the specific facial regions (eyes, jaw boundary, skin texture) that the model flags as manipulated — suitable for forensic report generation.
4. A complete modular Python codebase covering preprocessing, dual-stream inference, training pipeline, and evaluation scripts, uploaded to a public GitHub repository.
5. Full academic documentation: 60+ page project report, AKTU-compliant synopsis, and viva presentation slides.

4.2 Intended Users

- Digital forensic investigators and law enforcement agencies verifying video evidence.
- Banking and fintech platforms defending against deepfake-based KYC bypass attacks.
- Social media content moderation teams identifying and flagging manipulated media.
- Journalists and news organizations verifying the authenticity of political video content.

4.3 Future Scope

- Audio-visual synchronization analysis using lip-motion correlation and speech prosody features.
- Temporal inter-frame analysis using Video Transformers (TimeSformer) to detect blinking cadence anomalies.
- Mobile deployment via TensorRT quantization and ONNX Runtime for edge security cameras.
- Extension to detect diffusion-model generated content (Sora, Gen-2) and text-to-video synthesis.

5. Tentative Work Plan (10-Week Timeline)

Week / Timeline	Phase & Milestone	Key Activities & Deliverables
Week 1	Domain Finalisation & Problem Definition	Topic selection, mentor allotment (Dr. Devendra Gautam), preliminary literature search, dataset identification, problem formulation.
Week 2	Synopsis Preparation & Submission	Preparation of formal project synopsis in LIET-prescribed format. Submission for mentor review and approval.
Week 3	Synopsis Review & System Design	Mentor review and title approval. UML use-case design, DFD creation, finalization of dual-stream architecture diagram.
Week 4	Dataset Curation & Preprocessing Pipeline	FaceForensics++ dataset setup. Video frame extraction, MTCNN face detection, canonical 256×256 crop generation.
Week 5	Spatial Stream Implementation — Phase I	EfficientNet-B4 backbone integration via timm library. Training spatial-only baseline. Initial accuracy benchmarking.
Week 6	Mid-Term Review (≥30% Completion)	Formal mid-term evaluation: spatial pipeline demonstration, DCT spectral analysis prototype, progress log sign-off.
Week 7	Frequency Stream & Fusion Implementation	Full 2D-DCT transform + Spectral-CNN implementation. Attention-gated bilinear fusion module. Full DualStreamNet training.
Week 8	Integration, Testing & Dashboard Development	End-to-end pipeline integration. Grad-CAM implementation. Streamlit dashboard development. Cross-dataset testing on Celeb-DF v2.
Week 9	Report Writing & Code Cleanup	Complete project report drafting (60+ pages). GitHub repository finalization. Plagiarism check (Turnitin < 20%). Mentor corrections.
Week 10	Final Submission & Viva-Voce Defense	Spiral-bound report submission. Live demo of Streamlit dashboard. PowerPoint presentation. Panel question-and-answer defense.

6. Software / Hardware Required

6.1 Software Requirements

Software / Framework	Purpose / Version
Operating System	Windows 10/11 or Ubuntu 20.04 LTS
Programming Language	Python 3.10+
Deep Learning Framework	PyTorch 2.0+ with CUDA 11.8 (GPU) / CPU fallback
Model Library	timm (PyTorch Image Models) for EfficientNet-B4 pretrained weights
Signal Processing	SciPy (2D-DCT), NumPy (array operations)
Computer Vision	OpenCV 4.8+ (video decoding, frame extraction), MTCNN (face detection)
Web Dashboard	Streamlit 1.28+, Plotly 5.17+ (interactive gauges and charts)
Development IDE	VS Code / Jupyter Notebook
Version Control	Git + GitHub (repository: github.com/DevRanaKaila/DeepGuard-AI)

6.2 Hardware Requirements

Hardware Component	Minimum Specification
Processor	Intel Core i5 (8th Gen+) or AMD Ryzen 5 3500U or above
RAM	Minimum 8 GB DDR4 (16 GB recommended for training)
Storage	50 GB free SSD space (for datasets, models, and code)
GPU (Optional)	NVIDIA GPU with 4 GB VRAM+ for accelerated training (CUDA 11.8)
Network	Internet access for dataset download and package installation

7. References

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